

Activity No.6

Date: / /

Aim: To prepare 100 mL of 0.1 M standard solution of ferrous ammonium sulphate (Mohr's salt)

Theory: The molecular formula of ferrous ammonium sulphate is $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$.

Its molar mass is 392 g/mol.

1000 mL of 1 M solution of F.A.S. solution contains 392 g of $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$

Hence 100 mL of 0.1 M solution of F.A.S. will contain = $\frac{100 \times 0.1 \times 392}{1000 \times 1} = 3.92$ g of F.A.S.,

Apparatus: 100 mL volumetric flask, beaker, glass rod, balance watch glass, etc.

Chemicals: F.A.S. (Mohr's salt), distilled water, dilute H_2SO_4

Procedure:

1. Weigh accurately 3.92 g Ferrous ammonium sulphate on watch glass.
2. Transfer the weighed F.A.S. to a beaker and wash the watch glass with distilled water and transfer washing to the beaker.
3. Add little distilled water to dissolve it by stirring.
4. Transfer the solution of F.A.S. from beaker to 100 mL standard flask.
5. Wash the beaker twice with distilled water and transfer washing to the standard flask.
6. Dilute the solution up to the mark on standard flask to make volume 100 mL.
7. Add few drops of H_2SO_4 to get clear solution.

Result/Conclusion:

Remark and sign of teacher:

MCQ

Select [✓] the most appropriate answer from given alternatives of each sub question.

1. The molar mass of F.A.S. is----
 a. 392 u b. 392 g/mol c. 39.2 u d. 39.2 g/mol
2. To prepare standard 1 M solution of F.A.S.
 a. 0.392 g of F.A.S. of dissolved in distilled water and diluted to 1 L of solution
 b. 3.92 g of F.A.S. is dissolved in distilled water and diluted to 1 L of solution
 c. 39.2 g of F.A.S. is dissolved in distilled water and diluted to 1 L of solution
 ✓d. 392 g of F.A.S. is dissolved in distilled water and diluted to 1 L of solution
3. The second name for ferrous ammonium sulphate (F.A.S.) is
 a. Rochelle salt ✓b. Mohr's salt c. common salt d. sea salt
4. To prepare 100 mL of standard 0.1 M solution of F.A.S.
 a. 0.392 g of F.A.S. is dissolved in distilled water and diluted to 100 cm³
 ✓b. 3.92 g of F.A.S. is dissolved in distilled water and diluted to 100 cm³
 c. 39.2 g of F.A.S. is dissolved in distilled water and diluted to 100 cm³
 d. 392 g of F.A.S. is dissolved in distilled water and diluted to 100 cm³
5. The green colour of 0.1 M F.A.S. solution turns to which colour if we keep that solution open in an air for a long time?
 a. red b. blue ✓c. yellowish brown d. violet

Short answer questions

1. Calculate the amount of F.A.S. required to prepare 1000 mL of 0.1 M standard solution of F.A.S.,

$$\begin{aligned} \text{Molar mass of F.A.S.} &= 392 \text{ g mol}^{-1} \\ \text{Required mass} &= M \times \text{Molar mass} \times \text{Volume (L)} \\ &= 0.1 \times 392 \times 1 = 39.2 \text{ g} \end{aligned}$$

2. Calculate the molar mass of F.A.S. (Fe = 56, S = 32, N = 14, O = 16, H = 1)

$$\begin{aligned} \text{F.A.S.} &= \text{FeSO}(\text{NH})\text{SO} \cdot 6\text{H}_2\text{O} \\ &= 56 + 2(32) + 8(16) + 2(14) + 8(1) + 6(18) \\ &= 56 + 64 + 128 + 28 + 8 + 108 \\ &= 392 \text{ g mol}^{-1} \end{aligned}$$

3. Why the 0.1 M F.A.S. solution becomes yellowish brown on keeping open for a long time?

On exposure to air, ferrous ions (Fe^{2+}) in F.A.S. are oxidized to ferric ions (Fe^{3+}). Ferric ions impart a yellowish-brown colour to the solution.

4. Why it is necessary to add few drops of conc. H_2SO_4 in the preparation of F.A.S. solution?

A few drops of concentrated H_2SO_4 are added to prevent hydrolysis and oxidation of ferrous ions, thereby maintaining the stability of the F.A.S. solution.

5. How many water of crystallisation are present in Mohr's salt?

Mohr's salt contains 6 molecules of water of crystallisation ($6\text{H}_2\text{O}$).

Remark and sign of teacher: _____